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**GROUP 14**





## Eastern Syria Agro Climate/ Group14

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# Background & Objective



- **Brief project overview:** 20-year climate analysis (2006–2025) of NASA POWER data in Eastern Syria (Al-Hasakah).
- **Importance of the problem:** Severe droughts & heat stress directly threaten Syria's agricultural breadbasket and crop yields.
- **Project objectives:** Compute SPI drought indices and deploy an interactive FastAPI early-warning web app.

# SDG Relation



- **SDG 13 (Climate Action):** Strengthens climate resilience through SPI drought early-warning metrics
- **SDG 2 (Zero Hunger):** Protects rainfed wheat & barley crops to maintain regional food security.
- **SDG 15 (Life on Land):** Mitigates land degradation and water scarcity impacts in Eastern Syria.

# Solution / Project Element



- **Climate Data Engine:** Automated processing of 20-year precipitation and temperature data from NASA POWER.
- **SPI Drought Index:** Quantitative assessment categorizing historical drought cycles and water stress levels.
- **Interactive FastAPI & Cloud App:** Public web dashboard hosted on Vercel providing real-time drought alerts

# RoadMap



- **Current Phase (Completed):** 20-year climate analytics, SPI calculation, and live FastAPI deployment on Vercel.
- **Short-Term (3–6 Months):** Integrate Machine Learning (LSTM / XGBoost) for seasonal drought forecasting.
- **Long-Term (1 Year):** Incorporate Satellite GIS data (NDVI index) for real-time crop health monitoring.

# Project Development

1. **Data Collection & Processing:** Extracted and processed 20 years (2006–2025) of NASA POWER climate data using Python (Pandas/NumPy).
2. **SPI Drought Index Calculation:** Implemented Standardized Precipitation Index algorithms to identify drought severity levels.
3. **FastAPI & Platform Deployment:** Developed a lightweight, high-performance API and interactive web platform deployed on Vercel.

# Project Impact



1. **20+ Years Historical Database:** Provides a reliable climate baseline for Al-Hasakah to empower agricultural planning.
2. **Interactive Early Warning System:** Alerts farmers to drought risks early, optimizing supplemental irrigation timing.
3. **Open & Scalable Platform:** Delivers a fast, lightweight FastAPI solution offering free, real-time agricultural advisory.

# Conclusion and Futurework



- **Current Progress:**  
Successfully analyzed 20 years of NASA POWER climate data (2006–2025), calculated SPI drought indices, and deployed a live FastAPI platform on Vercel.
- **Next Phase (3–6 Months):**  
Integrate predictive machine learning models (LSTM / XGBoost) to forecast drought indices 3 to 6 months in advance.
- **Future Vision (1 Year):**  
Incorporate satellite remote sensing data (NDVI / Sentinel-2) for real-time crop health monitoring and soil moisture assessment across Al-Hasakah.



# Thank you!

